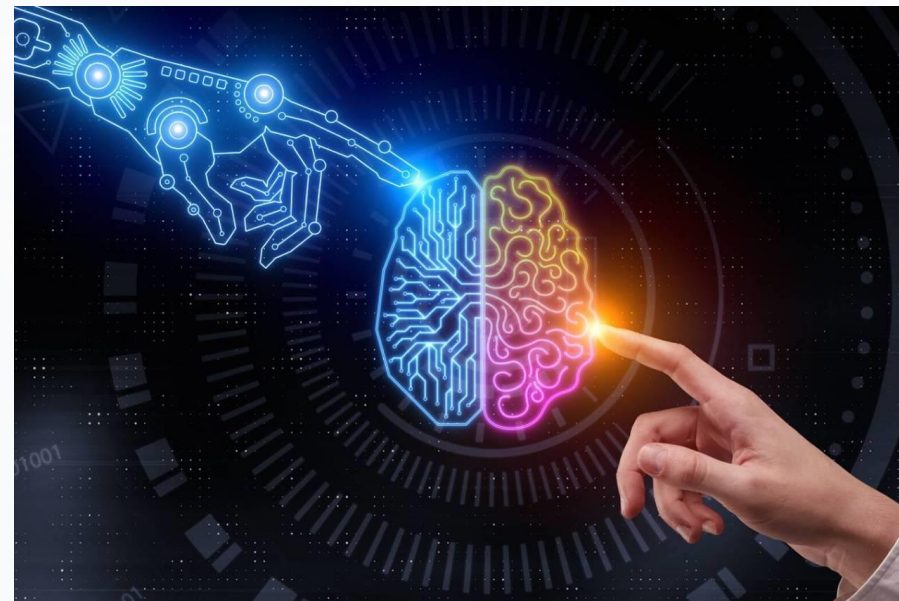




Fundamentos em Redes Neurais

Revisão de Álgebra Linear

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Sejam Bem-vindos !



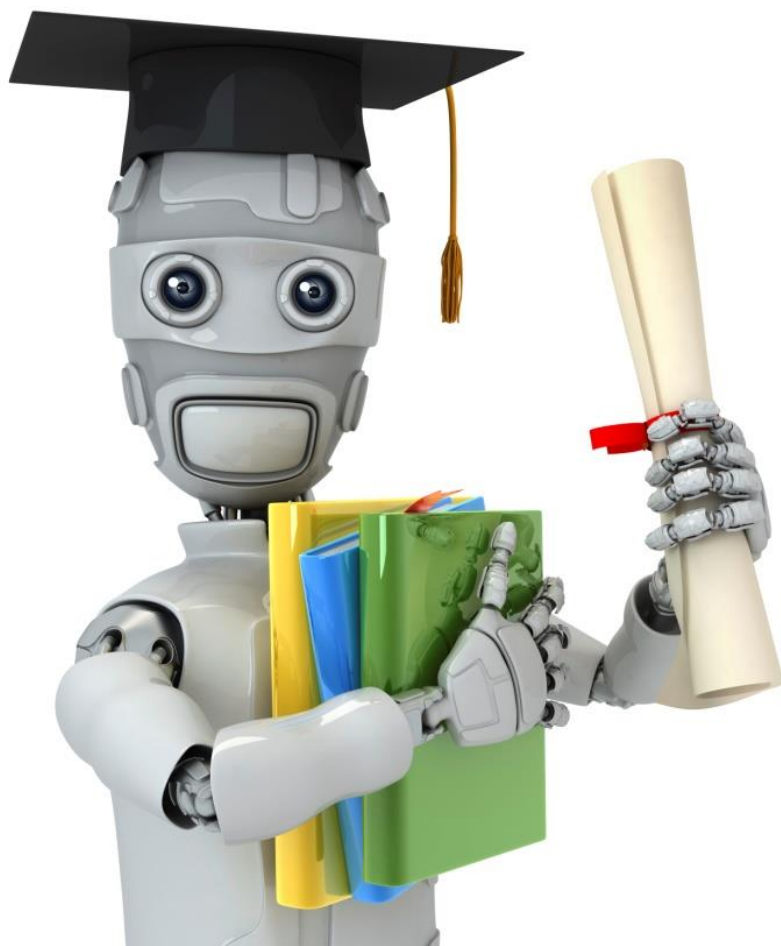
**Os celulares devem
ficar no silencioso
ou desligados**

Pode ser utilizado
apenas em caso
de emergência



**Boa tarde/noite, por
favor e com licença
DEVEM ser usados**

Educação é
essencial

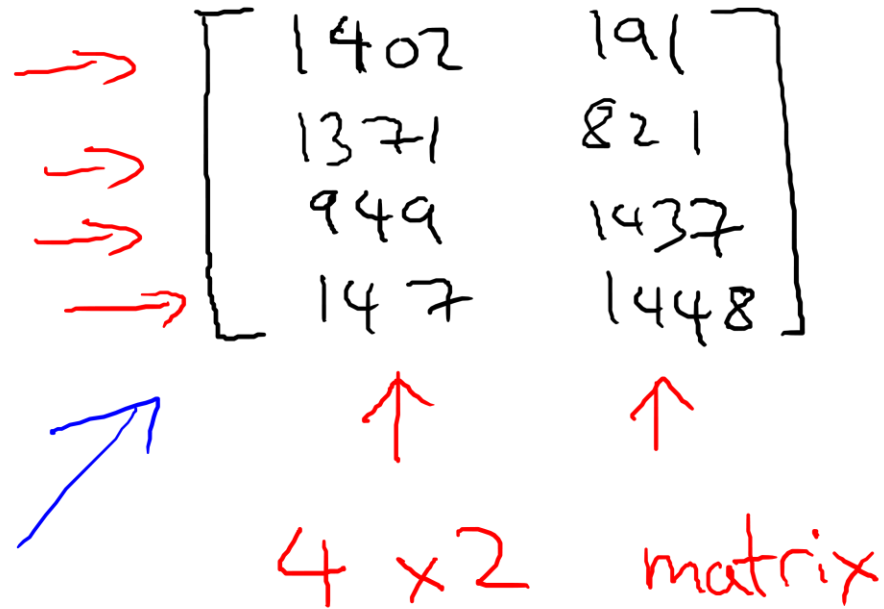


Machine Learning

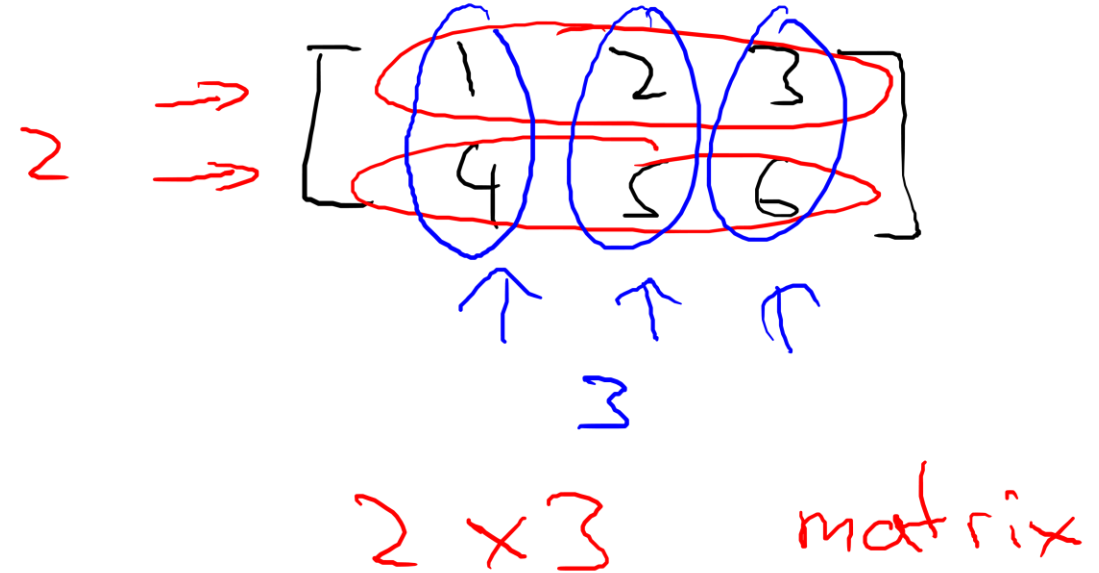
Linear Algebra
review (optional)

Matrices and
vectors

Matrix: Rectangular array of numbers:



→ $\mathbb{R}^{4 \times 2}$



$\mathbb{R}^{2 \times 3}$

Dimension of matrix: number of rows x number of columns

Matrix Elements (entries of matrix)

$$\underline{A} = \begin{bmatrix} \textcircled{1402} & \textcircled{191} \\ 1371 & 821 \\ 949 & \textcircled{1437} \\ \textcircled{147} & 1448 \end{bmatrix}$$

A_{ij} = " i, j entry" in the i^{th} row, j^{th} column.

$$A_{11} = 1402$$

$$A_{12} = 191$$

$$A_{32} = 1437$$

$$A_{41} = 147$$

$$\cancel{A_{43}} = \text{Undefined (error)}$$

Vector: An $n \times 1$ matrix.

$$\underline{y} = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$$

$n = 4$

← 4-dimensional vector.

~~$\mathbb{R}^{3 \times 2}$~~

$\mathbb{R}^4$

$y_i = i^{th}$ element

$$y_1 = 460$$

$$y_2 = 232$$

$$y_3 = 315$$

→ A, B, C, X

a, b, x, y

1-indexed vs 0-indexed:

$y[1]$

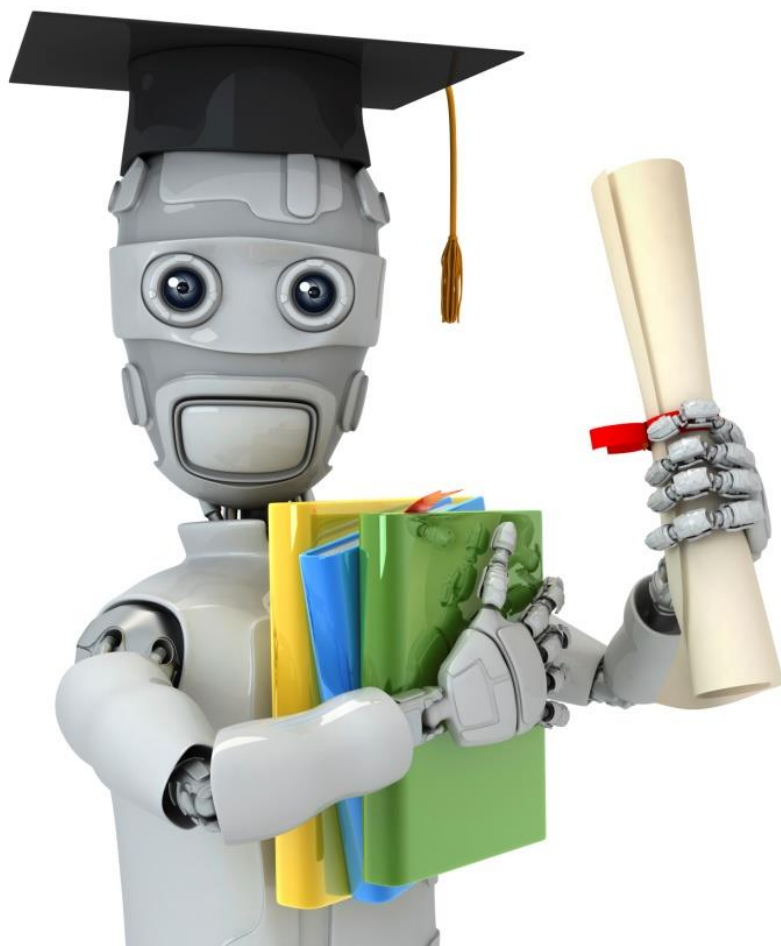
$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

1-indexed

$y[0]$

$$y = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

0-indexed



Machine Learning

Linear Algebra review (optional)

Addition and scalar multiplication

Matrix Addition

$$\begin{array}{c} \downarrow \downarrow \\ \rightarrow \left[\begin{array}{cc} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{array} \right] + \left[\begin{array}{cc} 4 & 0.5 \\ 2 & 5 \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} 5 & 0.5 \\ 4 & 10 \\ 3 & 2 \end{array} \right] \\ \text{3x2 matrix} \quad \text{3x2} \quad \text{3x2} \end{array}$$

$$\begin{array}{c} \rightarrow \left[\begin{array}{cc} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{array} \right] + \left[\begin{array}{cc} 4 & 0.5 \\ 2 & 5 \end{array} \right] = \text{error} \\ \text{3x2} \quad \text{2x2} \end{array}$$

Scalar Multiplication

← real number

$$3 \times \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 6 & 15 \\ 9 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} \times 3$$

3x2 3x2

$$\begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} / 4 = \frac{1}{4} \begin{bmatrix} 4 & 0 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{3}{2} & \frac{3}{4} \end{bmatrix}$$

Combination of Operands

Scalar multiplication

$$3 \times \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} / 3$$

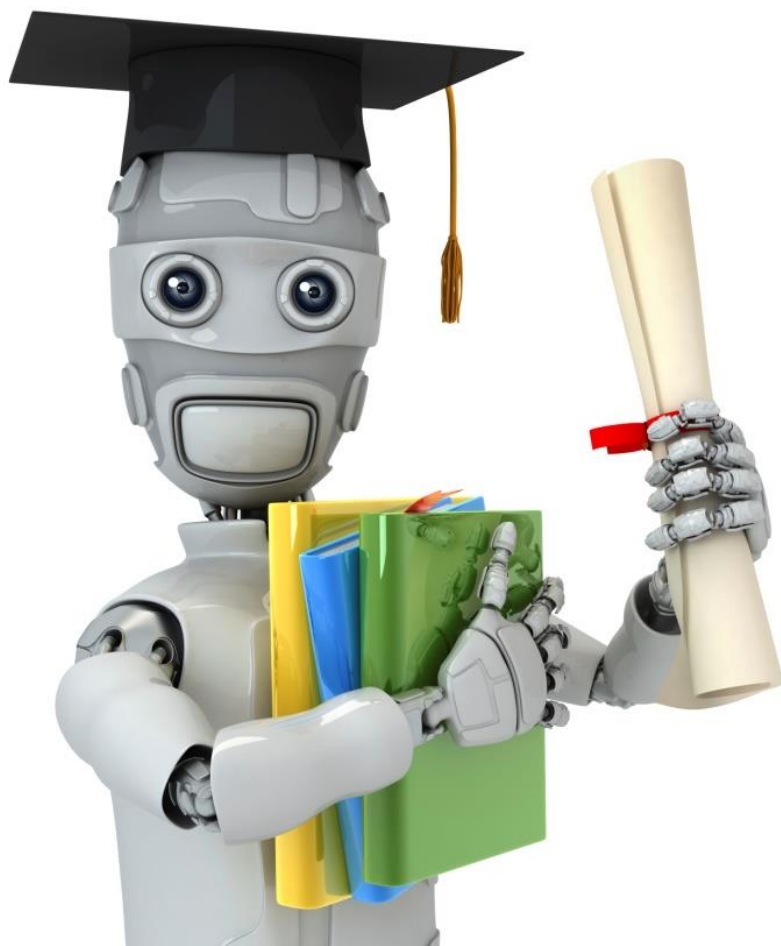
Scalar division

matrix subtraction / vector subtraction

matrix addition / vector addition

3x1 matrix
3-dimensional vector

$$= \begin{bmatrix} 3 \\ 12 \\ 6 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ \frac{2}{3} \end{bmatrix}$$
$$= \begin{bmatrix} 2 \\ 12 \\ 10\frac{1}{3} \end{bmatrix}$$



Machine Learning

Linear Algebra review (optional)

Matrix-vector multiplication

Example

$$\begin{bmatrix} 1 & 3 \\ 4 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 16 \\ 4 \\ 7 \end{bmatrix}$$

3×2 2×1 3×1 matrix

$$1 \times 1 + 3 \times 5 = 16$$

$$4 \times 1 + 0 \times 5 = 4$$

$$2 \times 1 + 1 \times 5 = 7$$

Details:

$$\begin{array}{ccc} \underline{A} & \times & \underline{x} \\ \begin{array}{c} \text{[Hand-drawn matrix with } m \text{ rows and } n \text{ columns, colored circles, and arrows indicating dimensions]} \end{array} & \times & \begin{array}{c} \text{[Hand-drawn vector with } n \text{ elements, colored circles, and arrows indicating dimension]} \end{array} \\ \begin{array}{c} \boxed{m} \times \boxed{n} \text{ matrix} \\ \text{(} m \text{ rows, } \\ \text{ } n \text{ columns)} \end{array} & & \begin{array}{c} \boxed{n} \times 1 \text{ matrix} \\ \text{(} n\text{-dimensional} \\ \text{vector)} \end{array} \end{array} = \underline{y} = \begin{array}{c} \text{[Hand-drawn vector with } m \text{ elements, colored circles, and arrows indicating dimension]} \\ \begin{array}{c} \boxed{m}\text{-dimensional} \\ \text{vector} \end{array} \end{array}$$

→ To get y_i , multiply A 's i^{th} row with elements of vector x , and add them up.

Example

$$\begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 3 & 0 & 4 \\ -1 & -2 & 0 & 0 \end{bmatrix}_{3 \times 4} \begin{matrix} \downarrow \\ 1 \\ 3 \\ 2 \\ 1 \end{matrix}_{4 \times 1} = \begin{bmatrix} 14 \\ 13 \\ -7 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 14 \\ 13 \\ -7 \end{bmatrix}$$

$$\left. \begin{array}{l} 1 \times 1 + 2 \times 3 + 1 \times 2 + 5 \times 1 = 14 \\ 0 \times 1 + 3 \times 3 + 0 \times 2 + 4 \times 1 = 13 \\ -1 \times 1 + (-2) \times 3 + 0 \times 2 + 0 \times 1 = -7 \end{array} \right\}$$

House sizes:

- 2104
- 1416
- 1534
- 852

Matrix

$$\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix}$$

4x2

$$h_{\theta}(x) = -40 + 0.25x$$

$h_{\theta}(x)$

2x1

Vector

$$\begin{bmatrix} -40 \\ 0.25 \end{bmatrix}$$

X

=

4x1 matrix

$$\begin{bmatrix} -40 \times 1 + 0.25 \times 2104 \\ -40 \times 1 + 0.25 \times 1416 \\ \\ \end{bmatrix}$$

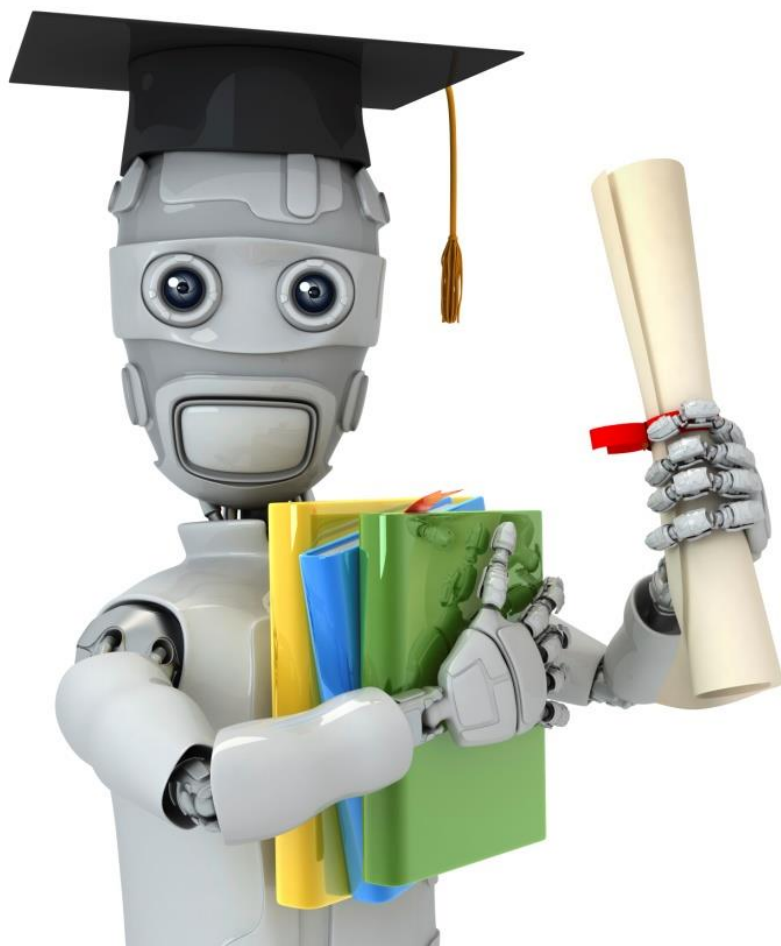
$h_{\theta}(2104)$

$h_{\theta}(1416)$

$$\text{prediction} = \text{Data Matrix} \times \text{Parameters}$$

4x1

for $i = 1:1000$,
prediction(i) = ...



Machine Learning

Linear Algebra review (optional)

Matrix-matrix multiplication

Example

$$\begin{array}{l} \begin{array}{c} \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 0 & 1 \\ \hline 5 & 2 \\ \hline \end{array} \\ \hline \textcircled{2 \times 3} \quad \textcircled{3 \times 2} \end{array} = \begin{bmatrix} 11 & 10 \\ 9 & 14 \end{bmatrix} \\ \begin{array}{c} \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{array}{|c|} \hline 1 \\ \hline 0 \\ \hline 5 \\ \hline \end{array} \\ \hline \end{array} = \begin{bmatrix} 11 \\ 9 \end{bmatrix} \\ \begin{array}{c} \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{array}{|c|} \hline 3 \\ \hline 1 \\ \hline 2 \\ \hline \end{array} \\ \hline \end{array} = \begin{bmatrix} 10 \\ 14 \end{bmatrix} \end{array}$$

Diagram illustrating matrix multiplication with dimensions and intermediate results:

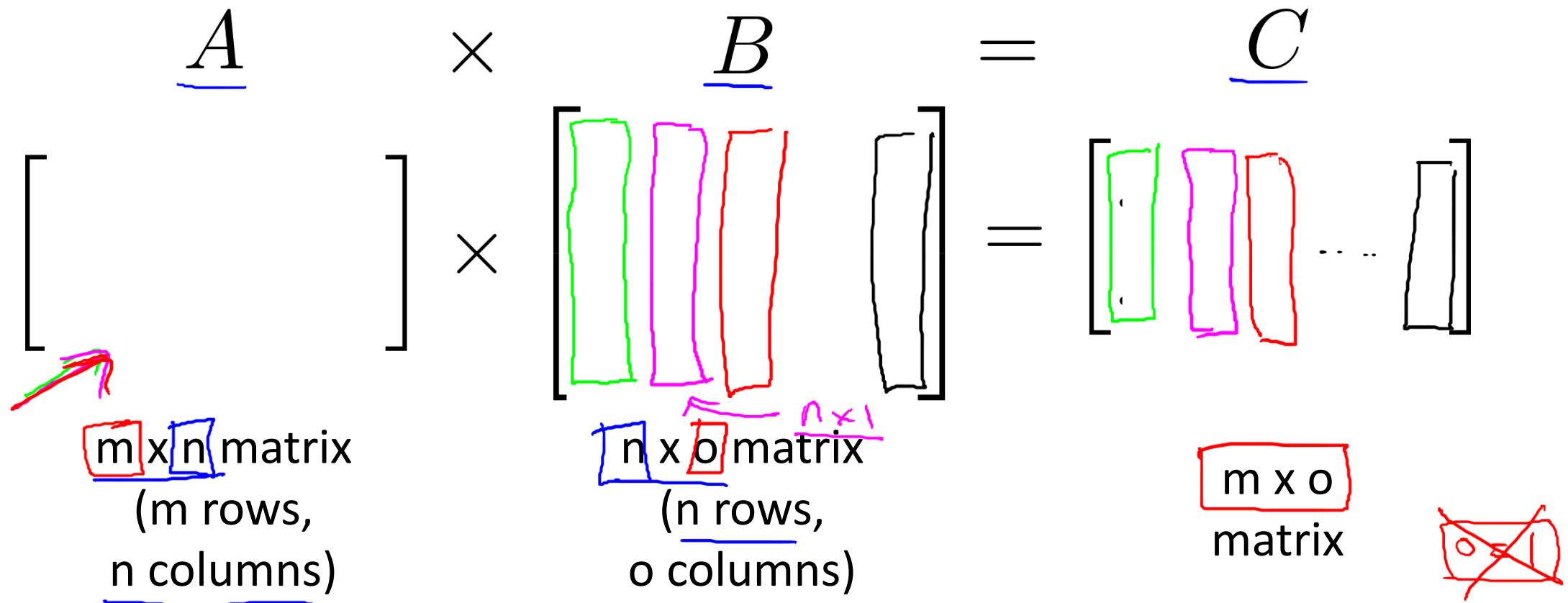
- Top row: $\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 11 & 10 \\ 9 & 14 \end{bmatrix}$. Dimensions 2×3 and 3×2 are indicated. The result matrix is shown with green and blue elements.
- Middle row: $\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 9 \end{bmatrix}$. The dimension 2×3 is indicated. The result matrix is shown with green elements.
- Bottom row: $\begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \end{bmatrix}$. The dimension 2×3 is indicated. The result matrix is shown with blue elements.

Arrows indicate the correspondence between the dimensions and the resulting matrix elements:

- Green arrows point from the green elements of the first matrix to the green elements of the first result matrix.
- Blue arrows point from the blue elements of the first matrix to the blue elements of the first result matrix.
- Green arrows point from the green elements of the second matrix to the green elements of the second result matrix.
- Blue arrows point from the blue elements of the second matrix to the blue elements of the second result matrix.

Details:

$$\begin{array}{c} \underline{A} \\ \left[\begin{array}{c} \end{array} \right] \\ \text{m} \times \text{n} \text{ matrix} \\ \text{(m rows,} \\ \text{n columns)} \end{array} \times \begin{array}{c} \underline{B} \\ \left[\begin{array}{c} \end{array} \right] \\ \text{n} \times \text{o} \text{ matrix} \\ \text{(n rows,} \\ \text{o columns)} \end{array} = \begin{array}{c} \underline{C} \\ \left[\begin{array}{c} \end{array} \right] \\ \text{m} \times \text{o} \\ \text{matrix} \end{array}$$



The i^{th} column of the matrix C is obtained by multiplying A with the i^{th} column of B . (for $i = 1, 2, \dots, o$)

Example

$$\overset{2 \times 2}{\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}} \overset{2 \times 2}{\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}} =$$

$$\begin{bmatrix} 9 & 7 \\ 15 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} =$$

$$\begin{bmatrix} 1 \times 0 + 3 \times 3 \\ 2 \times 0 + 5 \times 3 \end{bmatrix} =$$

$$\begin{bmatrix} 9 \\ 15 \end{bmatrix} \begin{bmatrix} 9 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} =$$

$$\begin{bmatrix} 1 \times 1 + 3 \times 2 \\ 2 \times 1 + 5 \times 2 \end{bmatrix} =$$

$$\begin{bmatrix} 7 \\ 12 \end{bmatrix}$$

House sizes:

$$\begin{cases} 2104 \\ 1416 \\ 1534 \\ 852 \end{cases}$$

Matrix

$$\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix}$$

Have 3 competing hypotheses:

1. $h_{\theta}(x) = -40 + 0.25x$

2. $h_{\theta}(x) = 200 + 0.1x$

3. $h_{\theta}(x) = -150 + 0.4x$

Matrix

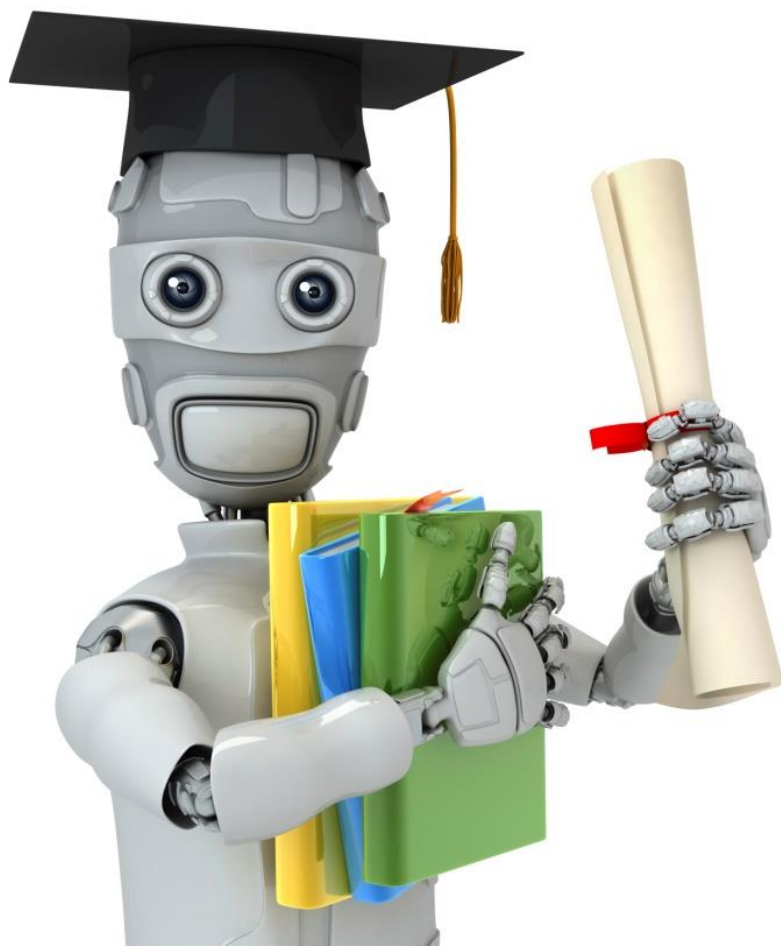
$$\begin{bmatrix} -40 & 0.25 \\ 200 & 0.1 \\ -150 & 0.4 \end{bmatrix}$$

=

$$\begin{bmatrix} 486 & 410 & 692 \\ 314 & 342 & 416 \\ 344 & 353 & 464 \\ 173 & 285 & 191 \end{bmatrix}$$

Prediction
of first
 h_{θ}

Predictions
of 2nd
 h_{θ}



Machine Learning

Linear Algebra review (optional)

Matrix multiplication properties

$$3 \times 5 = 5 \times 3$$


"Commutative"

Let A and B be matrices. Then in general,
 $A \times B \neq B \times A$. (not commutative.)

E.g.

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$


$$\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 2 \end{bmatrix}$$


|

$A \times B$
 $m \times n$

B
 $n \times m$

$A \times B$
 $m \times m$

B
 $n \times n$

$B \times A$
 $n \times n$

A
 $m \times m$




$$\underline{3 \times 5 \times 2}$$

$$3 \times 10 = 30 = 15 \times 2$$

$$3 \times (5 \times 2) = (3 \times 5) \times 2$$

"Associative"

$$\begin{array}{l} A \times (B \times C) \leftarrow \\ \underline{(A \times B)} \times C \leftarrow \end{array}$$

$$A \times B \times C.$$

Let $D = B \times C$. Compute $A \times D$.

Let $E = A \times B$. Compute $E \times C$.

$$A \times (B \times C)$$

$$(A \times B) \times C$$

Some
answer.

Identity Matrix

1 is identity.

$$1 \times z = z \times 1 = z$$

for any z

Denoted I (or $I_{n \times n}$).

Examples of identity matrices:

$$\begin{matrix} [1] \\ 1 \times 1 \end{matrix} \quad \begin{matrix} \boxed{1} & 0 \\ 0 & \boxed{1} \end{matrix} \\ 2 \times 2$$

$$\begin{matrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1} \end{matrix} \\ 3 \times 3$$

$$\begin{matrix} \boxed{1} & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & \boxed{1} \end{matrix} \\ 4 \times 4$$

Informally:

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

For any matrix A ,

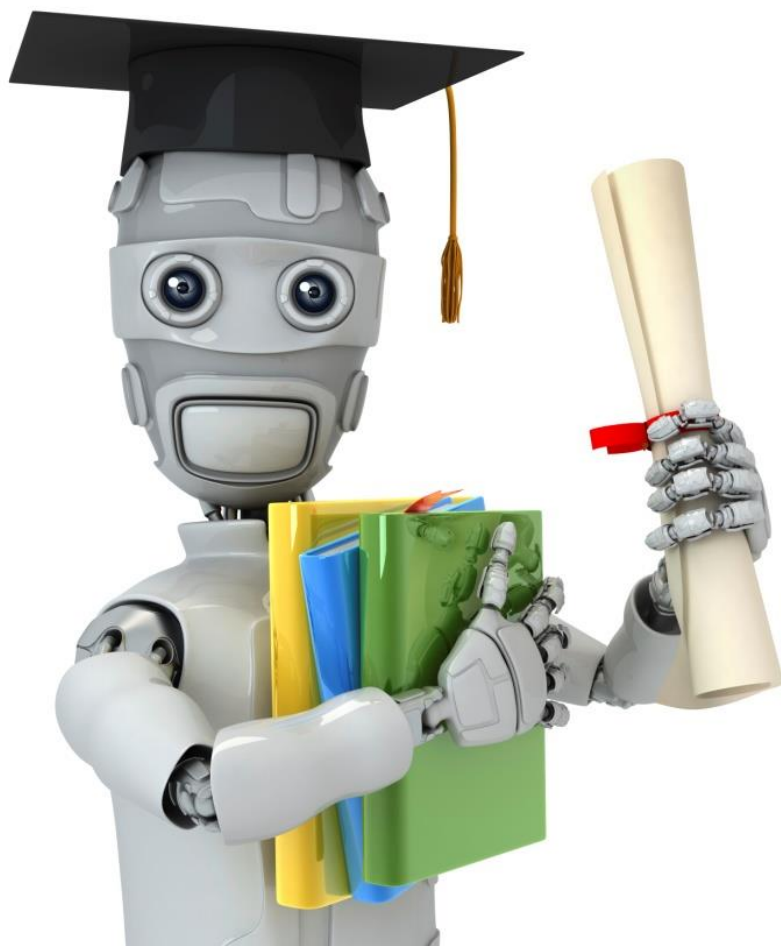
$$\begin{matrix} & \boxed{I} & = & \boxed{I} & \cdot & A & = & A \\ \uparrow & \uparrow & & \uparrow & & \uparrow & & \uparrow \\ m \times n & n \times n & & m \times m & & m \times n & & m \times n \end{matrix}$$

$$I_{n \times n}$$

Note:

$\underline{A} \underline{B} \neq \underline{B} \underline{A}$ in general

$$A I = \cancel{I A} I A \checkmark$$



Machine Learning

Linear Algebra
review (optional)

Inverse and
transpose

1 = "identity."

$$3 \underbrace{(3^{-1})}_{\frac{1}{3}} = 1$$

$$12 \times \underbrace{(12^{-1})}_{\frac{1}{12}} = 1$$

$$0 \underbrace{(0^{-1})}_{\text{undefined}}$$

Not all numbers have an inverse.

Matrix inverse:

square matrix
(# rows = # columns)

A^{-1}

If A is an $m \times m$ matrix, and if it has an inverse,

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\rightarrow \underline{A(A^{-1})} = \underline{A^{-1}A} = \underline{I}.$$

e.g.

$$\underbrace{\begin{bmatrix} 3 & 4 \\ 2 & 16 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 0.4 & -0.1 \\ -0.05 & 0.075 \end{bmatrix}}_{A^{-1}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{2 \times 2}$$

$A^{-1}A$

Matrices that don't have an inverse are "singular" or "degenerate"

Matrix Transpose

Example:

$$\underline{A} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 5 & 9 \end{bmatrix}$$

Handwritten annotations: A red arrow points to the underlined A. A green oval encloses the first row [1, 2, 0]. A blue oval encloses the second row [3, 5, 9]. A red '2 x 3' is written below the matrix. A blue arrow points from the first row to the first column of the transpose matrix.

$$\underline{B} = \underline{A}^T = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 0 & 9 \end{bmatrix}$$

Handwritten annotations: A blue arrow points to the underlined B. A green oval encloses the first column [1, 2, 0]. A blue oval encloses the second column [3, 5, 9]. A red '3 x 2' is written below the matrix.

Let A be an $m \times n$ matrix, and let $B = A^T$.

Then B is an $n \times m$ matrix, and

$$\underline{B}_{ij} = \underline{A}_{ji}.$$

$$B_{12} = A_{21} = 2$$

$$B_{32} = 9$$

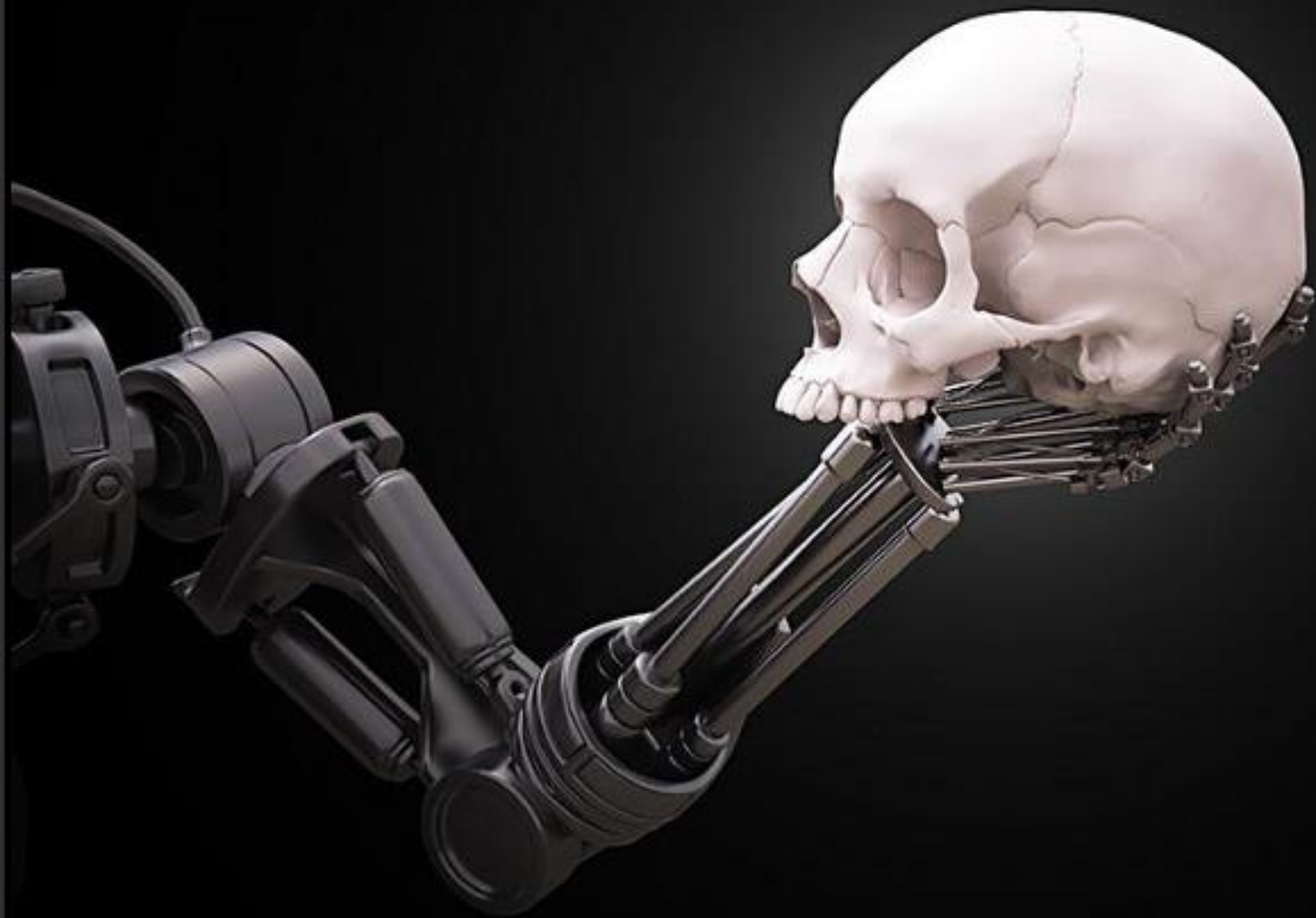
$$A_{23} = 9.$$

Referências

NG, Andrew. *Machine Learning*. Stanford University, 2011. Curso oferecido via Coursera. Disponível em: <https://www.coursera.org/learn/machine-learning>.



Dúvidas?



Até a próxima...



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